



Why Ethics?

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Potential Jobs for Computer and Data Scientists

Here is a list of potential jobs in the computer and data sciences.

Computer Science

- Software developer
 - Games
 - Social media
 - Airplane/Car navigation systems
 - Predictive algorithms (who is a good candidate for a loan or job, predicting future weather or erosion patterns, predicting if a person who is arrested is a flight risk)
 - Science applications (finding genetically-related diseases, analyzing tumor progression/reduction, drug discovery)
 - Compilers/operating systems
- Database Administrator
 - Design systems to store and retrieve important information
- Computer hardware engineer
 - Design the next phone, computer or tablet
 - Create/test circuit boards, routers, memory devices
 - Create ways for disabled to move and/or communicate
- Information security analyst
 - Detect cyber attacks
 - Prevent cyber attacks
- Web Developer
 - Create the technical structure for websites
 - Make sure that web pages are accessible and easily downloadable



- Structure sites to maximize the number of page views and visitors through search engine optimization.

Data Science and Statistics

- Data Analytics
 - Help develop strategies to make better business decisions
 - Create/use programs to help individuals invest money
 - Track and analyze website usage patterns
- Data architect
 - Find, clean and organize data
 - Look for patterns in data collected
- Machine learning
 - Develop predictive algorithms based on what has happened in the past (medicine, business, job/student success, economy, climate change, disease spread, loan repayment, criminal detainee flight risk etc.)
- Statistician
 - Interpret, analyze, and report statistical information (could be related to research in any field: business, psychology, sociology, medicine..)

Group work

In the next 10 minutes do the following in your group:

- Choose a note-taker and spokesperson so that your ideas can be shared with the larger group.
- Brainstorm with your group things that could go wrong in any of these areas if they are not handled carefully.
- Brainstorm with your group the things that could be beneficial to society if they were handled in the best possible way.

Example

A computer hardware designer could create a new phone that was feature rich but also designed to be obsolete within a year so that the user would need to replace it quickly. This would allow for a significant revenue stream for the company. However, the plethora of features could be too complex for an elderly person to manage, and the planned obsolescence could result in excessive waste.



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On the flip-side, the designer could implement a version of the phone where the initial setup included a choice of “essentials only” resulting in a screen that was easy to navigate and included a large, clean, display. There could be a prominent “help” button that connected the user with a predetermined relative or caregiver. The phone could come with an offer of a discount if the buyer’s previous phone was turned in for recycling (materials reuse) or donation to a non-profit.

Bibliography